

LTI System Estimation Using Least Squares

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Nathan Webb
Department of Electrical and Computer Engineering
University of Arizona
Tucson, AZ, USA

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1 Abstract

The purpose of this project was to delve into the theory, and the practical implementation of estimation for LTI (linear-time invariant) systems. By modeling an arbitrary LTI system as a discrete convolution of an input and the system's impulse response, it can be shown that this convolution is mathematically a linear transformation. This allows for the method of least squares to be used to approximate the impulse response of the system. A Python notebook was used to examine this theory and quantify the estimation accuracy under different independent variables such as signal output noise and input excitation level. The purpose of this program was to simulate real-world signal behaviors and to better understand how the theory behind least squares estimation could be used for practical applications.

2 Motivation

Estimation of a system is useful when the components of a system are unknown or inaccessible. In many practical scenarios this is the case. In such a situation a full analysis is impossible and thus the response of the system must be approximated via some algorithmic process.

This paper explores a method from linear algebra, “least squares”, and how an LTI system can inherently be modeled as a linear transformation. This concept allows for the application of the least squares method to help give a closest approximation of the system.

3 Theory

First we start with a fundamental LTI system concept, that the output of an LTI system can be expressed as the input convolved with the systems impulse response. This can be expressed as the following time-domain improper integral.

$$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

Where $x(t)$ is the input function, $h(t)$ is the impulse response of the systems and $y(t)$ is the output. Of course due to the integration, this convolution assumes infinite samples of both the input function. The integral is also defined from for times from negative infinity to positive infinity, which are bounds that do not make sense when working with discrete-time signals.

A discrete-time version of the convolution, over a finite sample can be expressed with the following sum.

$$y[n] = \sum_{k=0}^{N-1} h[k] x[n - k]$$

In this sum N is the number of signal samples, and n represents the n th sample. Because in a digital signal processing context, signals are non-continuous samples, this form of the convolution will be the most useful.

This form of the convolution does not make it immediately obvious that it is a linear transformation, however using the properties of dot products, a matrix can be derived which has the property of transforming the impulse response into the system output, making it mathematically equivalent to the convolution.

Lets define three vectors $\mathbf{x}$, $\mathbf{h}$ and $\mathbf{y}$.

$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \in \mathbb{R}^N$$
$$\mathbf{h} = \begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[K-1] \end{bmatrix} \in \mathbb{R}^K$$

$$\mathbf{y} = \begin{bmatrix} y[0] \\ y[1] \\ \vdots \\ y[N-1] \end{bmatrix} \in \mathbb{R}^N$$

Vector $\mathbf{x}$ contains samples from the input into an arbitrary system y . While $\mathbf{h}$ similarly contains samples of the impulse response of system y . Vector $\mathbf{y}$ contains elements of the systems output. Lets assume $K \leq N$, so to take the dot product between $\mathbf{x}$ and $\mathbf{h}$ we must take a subset of $\mathbf{x}$. Let say that for the n th sample of $\mathbf{x}$ we have a complementary vector $\hat{\mathbf{x}}$ defined-

$$\hat{\mathbf{x}} = \begin{bmatrix} x[n] \\ x[n+1] \\ \vdots \\ x[n-(K-1)] \end{bmatrix} \in \mathbb{R}^K$$

The vector $\hat{\mathbf{x}}$ contains a subset of $\mathbf{x}$ starting at index n and continuing to index $n - (K - 1)$. If $N - 1 < n - (K - 1)$ the vector can be assumed to be padded with zeros past index $n = N - 1$. Lets take the dot product between $\hat{\mathbf{x}}$ and $\mathbf{h}$.

$$\hat{\mathbf{x}} \cdot \mathbf{h} = x[n] * h[0] + x[n+1] * h[1] + \dots + x[n-(K-1)] * h[K-1]$$

Notice that this form is exactly the expanded form of the discrete-time convolution at a particular sample n . This fact demonstrates that the convolution for a single output sample can be rewritten exactly as a dot product between a subset of our sample vector and the impulse response vector.

Using this concept, we can generate a matrix to represent the convolution for all output samples. A mathematical object known as a Toeplitz matrix can be constructed that will allow us to extend this idea of dot product convolution to our entire output sample space.

$$\mathbf{X} = \begin{bmatrix} x[0] & 0 & 0 & \cdots & 0 \\ x[1] & x[0] & 0 & \cdots & 0 \\ x[2] & x[1] & x[0] & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x[N-1] & x[N-2] & x[N-3] & \cdots & x[N-K] \end{bmatrix}$$

Each row of this matrix corresponds to a particular of our input of length K .

With this matrix we can now rewrite our convolution in a standard linear form.

$$\mathbf{y} = \mathbf{X}\mathbf{h}$$

From this form we can use the concept of least squares to find a best approximation of $\mathbf{h}$ which we will label as the vector $\hat{\mathbf{h}}$.

$$\hat{\mathbf{h}} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}$$

The first thing to notice about this approximation is that the rank of $\mathbf{X}$ plays a large factor in the accuracy of the approximation. If $\text{rank}(\mathbf{X}) \ll \text{dim}(\mathbf{h})$, then $\mathbf{X}$ will not be able to transform $\mathbf{y}$ to a sufficient number of dimensions to well approximate $\mathbf{h}$. This leads to the idea of rank-deficiency, and how the linear independence of $\mathbf{X}$ is important in producing a good approximation.

DRAFTING: Need to talk about conditioning and how it is related, but not exactly the idea of rank deficiency

4 Methodology

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This section will highlight the concepts utilized in experiments and the rationale behind decisions made in the experiments.

5 Experiments

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Will contain an explanation of all of the experiments performed, and what kind of data is being measured.

6 Results

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Will show plots and other forms of data to express the results of the experiments, as long as some brief explanations

7 Discussion

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Will provide brief explanations of why the results are important and how they prove the validity of this method for real-world system identification.

8 Conclusion

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Will give a overview of what was covered in the paper, why it is important, and possibly some other addendum's.